BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. C.

Q2. Last In, First Out.

Q3. The array must be sorted.

Q4. B, before changing A.next to X.

You Do pages · Independent questions

Q1. Left child of 75 because 60>50 and 60<75.

Q2. 18 is the left child of 20, which is the right child of 15.

Q3. A sequence of sorted inserts can make it a chain of n nodes; each lookup may visit every node.

Exit Ticket · Model answer

False. An unbalanced BST may degenerate into a chain with O(n) lookup.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

RHS COMPUTING / OCR H446 / mixed LESSON 199 · DRAFT

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

LESSON 199 · Trees and binary search trees

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. In chain A→B→C, which node follows B?

Q2. What does LIFO mean?

Q3. What must be true before binary search on an array?

Q4. Insert X between A and B: what should new X.next point to?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

root — Topmost tree node.

child — Node directly below a parent.

subtree — A node and all its descendants.

binary search tree — A binary tree ordered by values, with smaller keys to the left and greater to the right.

balanced — Height kept relatively small compared with node count.

New learning

Start at root — Compare the new value 25 with root 40; it belongs in the left subtree.

Compare again — 25 is greater than 20, so move right. The right child currently stores 30.

Insert without breaking the property — 25 is smaller than 30; insert it as the left child of 30.

Worked example

Insert 12 into BST with root 20, left 10 and right 30.

1. Predict — Compare 12 with 20 → go left to 10.

2. Confirm and explain — 12 > 10, so 12 becomes the right child of 10.

We Do · Practise with guidance

Insert 35 into BST with root 25, left 15, right 40.

```
35 > 25 → RIGHT
35 < 40 → LEFT
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Predict

35 is greater than 25, so go right.

Confirm and explain

35 is smaller than 40, so insert it as the left child of 40.

RHS COMPUTING / OCR H446 / mixed LESSON 199 · DRAFT

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

BST root 50, left child 25, right child 75: where does 60 go?

Question 2

Insert 15, 10, 20, 18 into an empty BST. Identify the parent of 18.

Question 3

Explain why a BST can have O(n) worst-case search time despite O(log n) average search on a reasonably balanced tree.

Exit Ticket · Answer before checking

Every binary search tree has logarithmic worst-case lookup time.